# The Root-System Extension of Ramanujan's Master Theorem Was Answered by arXiv:1211.0024

Literature-status note for arXiv:1103.5126
Run `fa_banach_001`, lane 6

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**Status: literature already answered.** The later paper is by the same authors, explicitly identifies the earlier symmetric-space theorem, and proves the requested root-system extension for arbitrary positive multiplicities.

## Source question

Gestur Ólafsson and Angela Pasquale, *Ramanujan's Master Theorem for Riemannian symmetric spaces*, arXiv:1103.5126 [1], ask in Section 8, "Further remarks and open problems," for a generalization to the Heckman–Opdam hypergeometric setting: after recalling the extension of compact and noncompact spherical transforms to hypergeometric functions associated with root systems, they call a root-system version of Ramanujan's Master Theorem a natural question. They single out even multiplicity as a case where shift operators should reduce the problem to a Euclidean $W$-invariant theorem. This passage is on page 33 of the PDF rebuilt from the cached arXiv source.

## Supporting answer

The same authors subsequently published *Ramanujan's Master theorem for the hypergeometric Fourier transform on root systems*, arXiv:1211.0024 [2]. Its abstract states that the paper proves precisely such an analogue and that it generalizes the earlier symmetric-space result to arbitrary positive multiplicity functions.

The exact result is Theorem 5.1, "Ramanujan's Master Theorem for root systems," on PDF pages 18–19. It starts with a root system $\Sigma$ and a positive multiplicity function $m$, forms the alternating normalized Heckman–Opdam Jacobi series from a Hardy-class coefficient function, proves its holomorphic convergence, represents its continuation by the hypergeometric Fourier inversion integral, and identifies its hypergeometric Fourier transform. Section 6 proves the theorem. Both the introduction and the proof explicitly describe it as the extension of the earlier Riemannian-symmetric-space theorem.

Thus arXiv:1211.0024 gives an author-known direct answer, and it is stronger than the even-multiplicity subcase suggested in arXiv:1103.5126.

## Scope and evidence note

This identification answers only the root-system extension. The source's separate broad remarks about higher-rank Bertram kernels and Paley–Wiener characterization remain outside this status claim.

The official supporting PDF download was unavailable because the environment's external-usage quota was exhausted. The official arXiv abstract and primary full-PDF text were inspected online, including Theorem 5.1 and Section 6; exact links and locations are recorded in `supporting_evidence.md`. The local `source_paper.pdf` was compiled from the cached official arXiv TeX source.

## References


[1] G. Ólafsson and A. Pasquale, *Ramanujan's Master Theorem for Riemannian symmetric spaces*, J. Funct. Anal. 262 (2012), 2151–2180; arXiv:1103.5126. `https://arxiv.org/abs/1103.5126`.

[2] G. Ólafsson and A. Pasquale, *Ramanujan's Master theorem for the hypergeometric Fourier transform on root systems*, arXiv:1211.0024 (2012). `https://arxiv.org/abs/1211.0024`.